

EXPERIMENT – 4 Using the I2C Bus

PROJECT NAME

Temperature Data Logger Using Arduino UNO, DHT11 Sensor, DS1307 RTC Module and SSD1306 OLED Display

PROBLEM STATEMENT

To design and implement a real-time temperature monitoring and recording system using Arduino UNO. The system continuously reads temperature using a DHT11 sensor and records temperature values every 15 minutes.

The DS1307 RTC module provides accurate real-time timestamps, while the SSD1306 OLED display shows live temperature and time information. The recorded temperature data is stored in Arduino EEPROM memory and transferred to a laptop through USB serial communication.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
DHT11 Temperature Sensor	1
DS1307 RTC Module	1
SSD1306 OLED Display	1
Jumper Wires	Required
Breadboard	1
USB Cable	1

PIN CONNECTIONS

DHT11 SENSOR CONNECTIONS

DHT11 Pin	Arduino Pin
DATA	D2
VCC	5V
GND	GND

DS1307 RTC CONNECTIONS

RTC Pin	Arduino Pin
SDA	A4
SCL	A5
VCC	5V
GND	GND

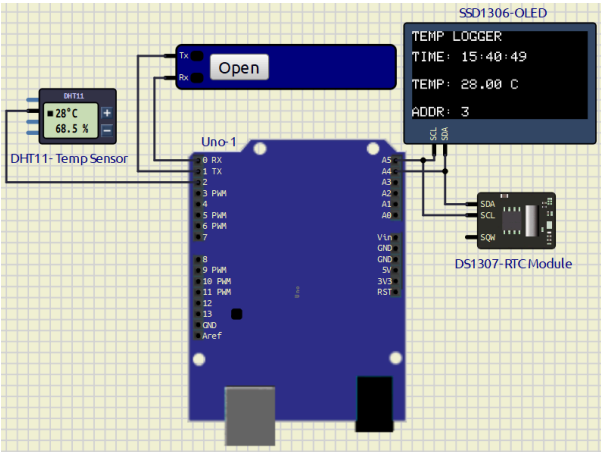
SSD1306 OLED CONNECTIONS

OLED Pin	Arduino Pin
SDA	A4
SCL	A5
VCC	5V
GND	GND

SOFTWARE USED

- Arduino IDE
- SimulIDE

Circuit Diagram



WORKING PRINCIPLE

1. DHT11 continuously measures temperature.
2. DS1307 RTC provides real-time clock information.
3. Arduino UNO reads:
 - Temperature
 - Current time

4. OLED display shows:
 - Current Time
 - Current Temperature
 - EEPROM Address
 5. Every 15 minutes:
 - Temperature value is stored into EEPROM.
 - Data is transferred to laptop through USB serial communication.
 6. Serial monitor displays recorded temperature logs.
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LIBRARIES USED

```
#include <Wire.h>
#include <RTClib.h>
#include <EEPROM.h>
#include <DHT.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

MAIN ARDUINO CODE

Complete Code With Comments----->[Click here](#)

```
#include <Wire.h>
#include <RTClib.h>
#include <EEPROM.h>
#include <DHT.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  -1
);

RTC_DS1307 rtc;

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

int eepromAddress = 0;
```

```
int lastRecordedMinute = -1;

void setup()
{
  Serial.begin(9600);

  Wire.begin();

  rtc.begin();

  dht.begin();

  // rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    while (1);
  }

  display.clearDisplay();
  display.display();

  Serial.println("TEMP LOGGER STARTED");
}

void loop()
{
  DateTime now = rtc.now();

  float temperature = dht.readTemperature();

  displayData(now, temperature);

  if (
    now.minute() % 15 == 0 &&
    now.minute() != lastRecordedMinute
  )
  {
    EEPROM.write(eepromAddress++, now.hour());

    EEPROM.write(eepromAddress++, now.minute());

    EEPROM.write(eepromAddress++, (int)temperature);

    Serial.print(now.hour());

    Serial.print(":");

    print2digitSerial(now.minute());

    Serial.print(":");

    print2digitSerial(now.second());
  }
}
```

```
        Serial.print(",");

        Serial.print(temperature);

        Serial.println(" C");

        lastRecordedMinute = now.minute();
    }

    delay(1000);
}

void displayData(DateTime now, float temperature)
{
    display.clearDisplay();

    display.setTextSize(1);

    display.setTextColor(WHITE);

    display.setCursor(0, 0);
    display.println("TEMP LOGGER");

    display.setCursor(0, 15);

    display.print("TIME: ");

    print2digit(now.hour());

    display.print(":");

    print2digit(now.minute());

    display.print(":");

    print2digit(now.second());

    display.setCursor(0, 35);

    display.print("TEMP: ");

    display.print(temperature);

    display.println(" C");

    display.setCursor(0, 55);

    display.print("ADDR: ");

    display.print(eepromAddress);

    display.display();
}
```

```
void print2digit(int number)
{
    if (number < 10)
    {
        display.print("0");
    }

    display.print(number);
}

void print2digitSerial(int number)
{
    if (number < 10)
    {
        Serial.print("0");
    }

    Serial.print(number);
}
```

OLED OUTPUT

TEMP LOGGER

TIME: 12:15:00

TEMP: 25 C

ADDR: 3

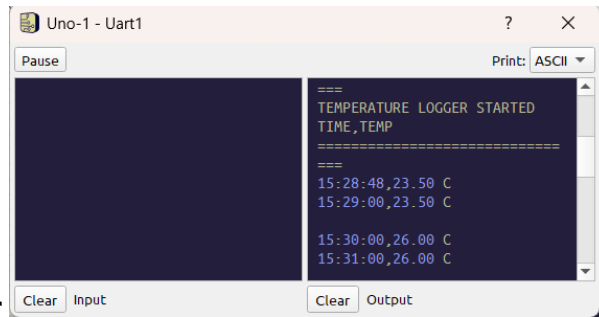
SERIAL OUTPUT

12:00:00,25 C

12:01:00,26 C

12:02:00,27 C

Instead of record data each 15min , I record it each 1 min onwards. so this is how its recorded and



displays in serial monitor

FEATURES

- Real-time temperature monitoring
 - Temperature recording every 15 minutes
 - OLED display output
 - RTC-based time stamping
 - EEPROM data storage
 - USB serial data transfer
 - Laptop data recording support
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ADVANTAGES

- No SD card required
 - Low-cost implementation
 - Easy USB communication
 - Real-time monitoring
 - Permanent EEPROM storage
 - Simple embedded system design
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APPLICATIONS

- Weather monitoring systems
 - Laboratory temperature logging
 - Industrial monitoring systems
 - Greenhouse monitoring
 - Environmental monitoring
 - Embedded system experiments
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CONCLUSION

The Temperature Data Logger System using Arduino UNO, DHT11 sensor, DS1307 RTC module, and SSD1306 OLED display was successfully implemented and tested. The system accurately records temperature every 15

minutes and transfers the data to a laptop through USB serial communication.

During this experiment, I learned:

- RTC interfacing
- OLED interfacing
- EEPROM memory handling
- Serial communication
- Temperature sensing
- Data logging techniques in embedded systems

REFERENCES

1. Arduino UNO Official Documentation
<https://docs.arduino.cc/hardware/uno-rev3/>
 2. DHT11 Sensor Documentation
<https://components101.com/sensors/dht11-temperature-sensor>
 3. DS1307 RTC Datasheet
<https://datasheets.maximintegrated.com/en/ds/DS1307.pdf>
 4. RTCLib Library
<https://github.com/adafruit/RTCLib>
 5. Adafruit SSD1306 Library
https://github.com/adafruit/Adafruit_SSD1306
 6. EEPROM Library Documentation
<https://www.arduino.cc/en/Reference/EEPROM>
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